



Faculty of Computers and Data Science

Cyber Security Program

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Security of distributed system – Task2

1. Valid JWT Token Test

For HTTP:

```
TOKEN=$(python3 get_token.py)
```

```
echo "Token: $TOKEN"
```

```
curl -s -H "Authorization: Bearer $TOKEN" http://localhost:8080/task | python3 -m json.tool
```

For HTTPS:

```
TOKEN=$(python3 get_token.py)
```

```
curl -k -H "Authorization: Bearer $TOKEN" https://localhost/task | python3 -m json.tool
```

[illegible]

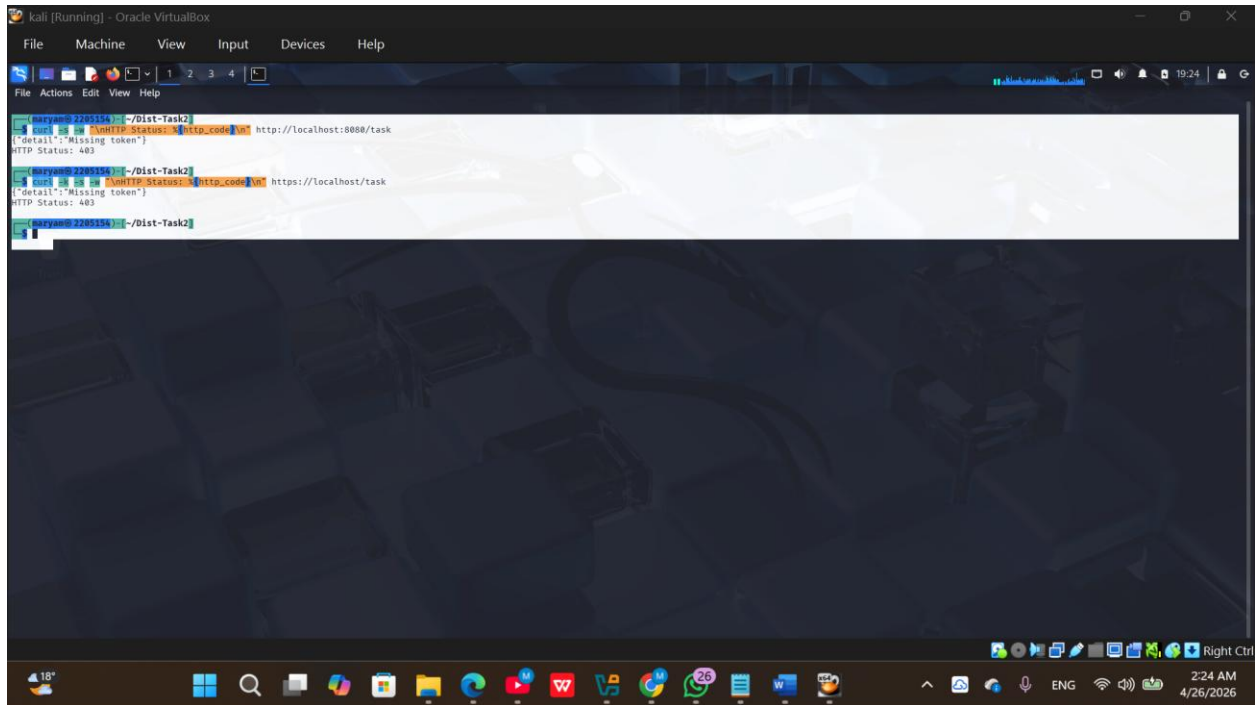
2. Missing Token Test

For HTTP:

```
curl -s -w "\nHTTP Status: %{http_code}\n" http://localhost:8080/task
```

For HTTPS:

```
curl -k -s -w "\nHTTP Status: %{http_code}\n" https://localhost/task
```



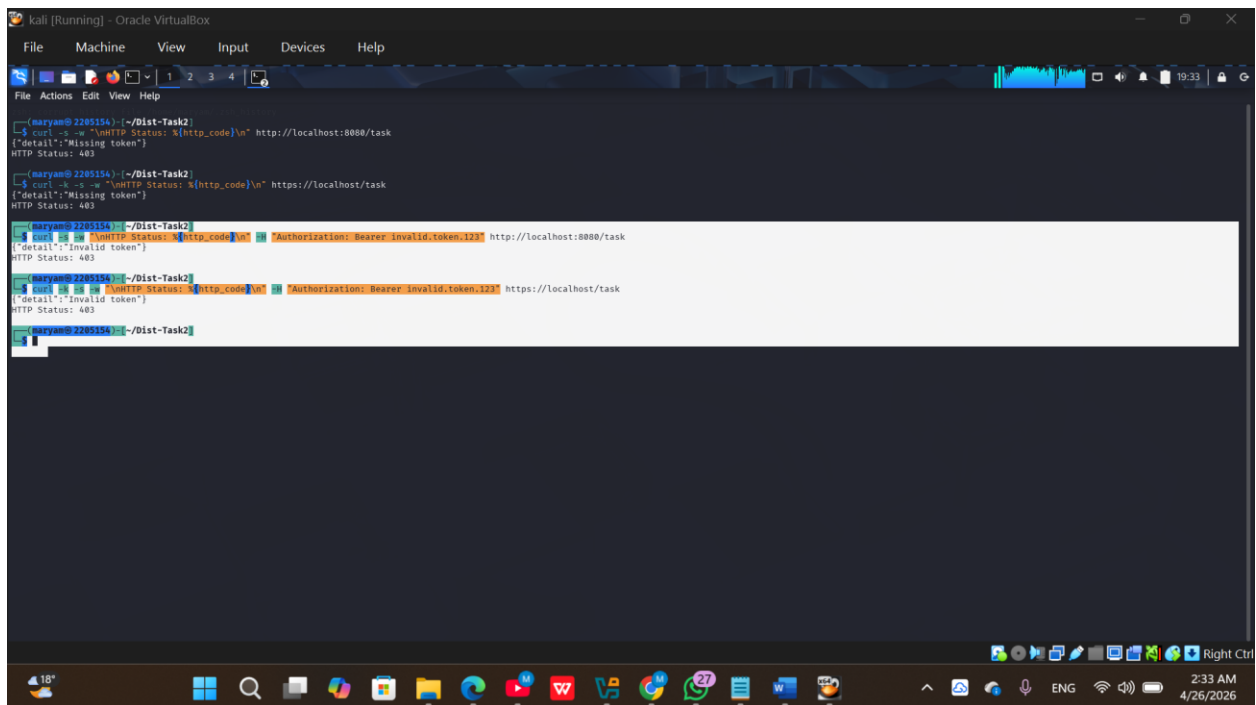
3. Invalid Token Test

For HTTP:

```
curl -s -w "\nHTTP Status: %{http_code}\n" -H "Authorization: Bearer invalid.token.123"  
http://localhost:8080/task
```

For HTTPS:

```
curl -k -s -w "\nHTTP Status: %{http_code}\n" -H "Authorization: Bearer invalid.token.123"  
https://localhost/task
```



```
kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
File Actions Edit View Help

(maryam@2205154) ~/Dist-Task2
$ curl -s -w "\nHTTP Status: %{http_code}\n" http://localhost:8080/task
{"detail": "Missing token"}
HTTP Status: 403

(maryam@2205154) ~/Dist-Task2
$ curl -k -s -w "\nHTTP Status: %{http_code}\n" https://localhost/task
{"detail": "Missing token"}
HTTP Status: 403

(maryam@2205154) ~/Dist-Task2
$ curl -s -w "\nHTTP Status: %{http_code}\n" -H "Authorization: Bearer invalid.token.123" http://localhost:8080/task
{"detail": "Invalid token"}
HTTP Status: 403

(maryam@2205154) ~/Dist-Task2
$ curl -k -s -w "\nHTTP Status: %{http_code}\n" -H "Authorization: Bearer invalid.token.123" https://localhost/task
{"detail": "Invalid token"}
HTTP Status: 403

(maryam@2205154) ~/Dist-Task2
```

4. RabbitMQ Queue Test

```
curl -s -u guest:guest http://localhost:15672/api/queues | python3 -c "
```

```
import sys,json
```

```
queues = json.load(sys.stdin)
```

```
for q in queues:
```

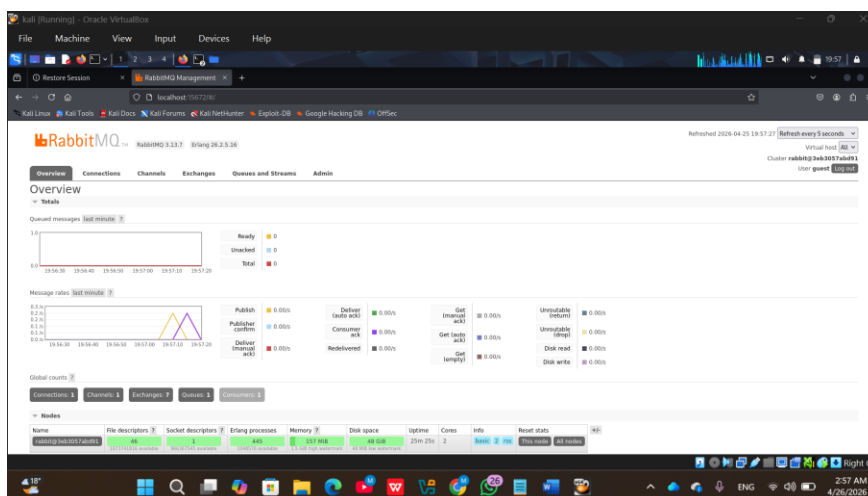
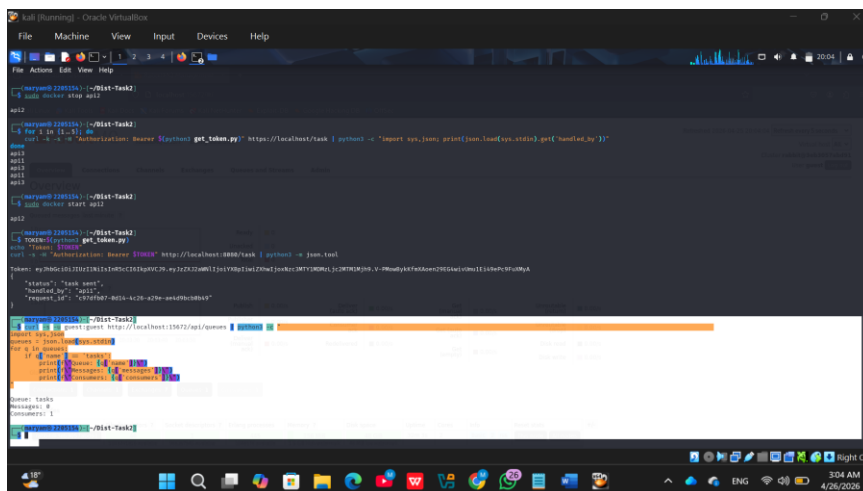
```
if q['name'] == 'tasks':
```

```
print(f"Queue: {q['name']}\n")
```

```
print(f"Messages: {q['messages']}")
```

```
print(f"Consumers: {q['consumers']}\n")
```

11



5. Request ID Tracking Test

Send request and capture Request ID

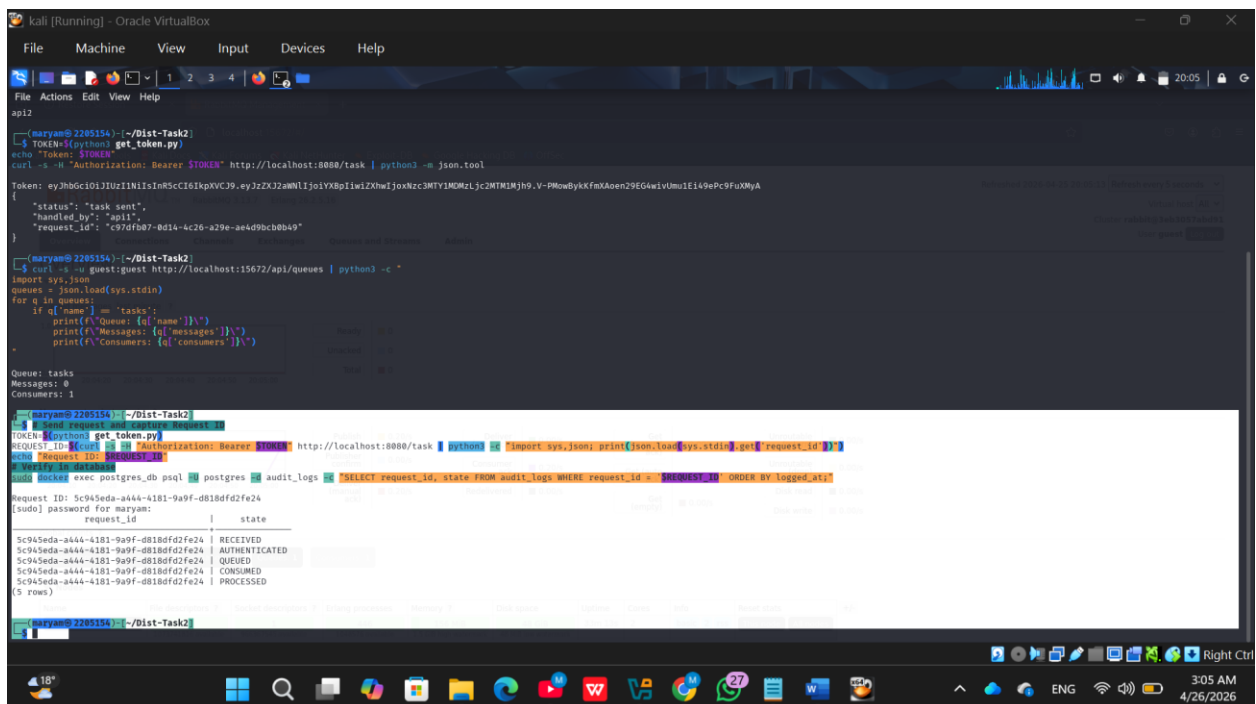
```
TOKEN=$(python3 get_token.py)
```

```
REQUEST_ID=$(curl -s -H "Authorization: Bearer $TOKEN" http://localhost:8080/task | python3 -c "import sys,json; print(json.load(sys.stdin).get('request_id'))")
```

```
echo "Request ID: $REQUEST_ID"
```

Verify in database

```
sudo docker exec postgres_db psql -U postgres -d audit_logs -c "SELECT request_id, state FROM audit_logs WHERE request_id = '$REQUEST_ID' ORDER BY logged_at;"
```



```
kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help
api2
(maryam@2205154) ~/Dist-Task2
$ TOKEN=$(python3 get_token.py)
echo "Token: $TOKEN"
Token: eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ2ZXJ2aWVudCI6IjoiYXJpIiwiaWF0IjoxNjY1MDMzLjY2MTI1MjM5LjV-Pk0wByKfXmXoEn29G4wlvUmu1E149ePc9FuXhYkA
$ curl -s -H "Authorization: Bearer $TOKEN" http://localhost:8080/task | python3 -m json.tool
{
  "status": "task sent",
  "handled_by": "api1",
  "request_id": "c97df0a7-8d14-4c26-a29e-ae4d9bc8b049"
}
(maryam@2205154) ~/Dist-Task2
$ curl -s -u guest:guest http://localhost:15672/api/queues | python3 -c "
import sys,json
queues = json.load(sys.stdin)
for q in queues:
    if q['name'] == 'tasks':
        print(f'\Queue: {q['name']}\n')
        print(f'\Messages: {q['messages']}\n')
        print(f'\Consumers: {q['consumers']}\n')
"
Queue: tasks
Messages: 0
Consumers: 1
(maryam@2205154) ~/Dist-Task2
$ echo "Request ID: $(curl -s -H "Authorization: Bearer $TOKEN" http://localhost:8080/task | python3 -c "import sys,json; print(json.load(sys.stdin).get('request_id'))")"
Request ID: c97df0a7-8d14-4c26-a29e-ae4d9bc8b049
(maryam@2205154) ~/Dist-Task2
$ sudo docker exec postgres_db psql -U postgres -d audit_logs -c "SELECT request_id, state FROM audit_logs WHERE request_id = '$REQUEST_ID' ORDER BY logged_at;"
Request ID: 5c945eda-a444-4181-9a9f-d818fd2fe24
[sudo] password for maryam:
request_id | state
-----
5c945eda-a444-4181-9a9f-d818fd2fe24 | RECEIVED
5c945eda-a444-4181-9a9f-d818fd2fe24 | AUTHENTICATED
5c945eda-a444-4181-9a9f-d818fd2fe24 | QUEUED
5c945eda-a444-4181-9a9f-d818fd2fe24 | CONSUMED
5c945eda-a444-4181-9a9f-d818fd2fe24 | PROCESSED
(5 rows)
```

6. Load Balancing Test:

For HTTP:

for i in {1..15}; do

echo -n "Request \$i: "

curl -s -H "Authorization: Bearer \$(python3 get_token.py)" http://localhost:8080/task |
python3 -c "import sys,json; print(json.load(sys.stdin).get('handled_by'))"

done

For HTTPS:

for i in {1..15}; do

echo -n "Request \$i: "

curl -k -s -H "Authorization: Bearer \$(python3 get_token.py)" https://localhost/task | python3
-c "import sys,json; print(json.load(sys.stdin).get('handled_by'))"

done

```
kali [Running] - Oracle VirtualBox
File Machine View Input Devices Help

File Actions Edit View Help
04bd4287-9d04-4f08-97df-76b67ba71f5f | worker | PROCESSED | | 2026-04-25 22:29:02.474176
84bd4287-9d04-4f08-97df-76b67ba71f5f | worker | CONSUMED | | 2026-04-25 22:29:02.456771
(15 rows)

(maryam@2205154) ~/Dist-Task2
for i in {1..15}; do
  echo -n "Request $i: "
  curl -s -H "Authorization: Bearer $(python3 get_token.py)" http://localhost:8080/task | python3 -c "import sys,json; print(json.load(sys.stdin).get('handled_by'))"
done
Request 1: ap11
Request 2: ap12
Request 3: ap13
Request 4: ap11
Request 5: ap12
Request 6: ap13
Request 7: ap11
Request 8: ap12
Request 9: ap13
Request 10: ap11
Request 11: ap12
Request 12: ap13
Request 13: ap11
Request 14: ap12
Request 15: ap13

(maryam@2205154) ~/Dist-Task2
for i in {1..15}; do
  echo -n "Request $i: "
  curl -k -s -H "Authorization: Bearer $(python3 get_token.py)" https://localhost/task | python3 -c "import sys,json; print(json.load(sys.stdin).get('handled_by'))"
done
Request 1: ap11
Request 2: ap12
Request 3: ap13
Request 4: ap11
Request 5: ap12
Request 6: ap13
Request 7: ap11
Request 8: ap12
Request 9: ap13
Request 10: ap11
Request 11: ap12
Request 12: ap13
Request 13: ap11
Request 14: ap12
Request 15: ap13
```

7. Rate Limiting Test

For HTTP:

```
echo "Sending 20 rapid requests..."
```

```
for i in {1..20}; do
```

```
curl -s -o /dev/null -w "%{http_code}\n" -H "Authorization: Bearer $(python3 get_token.py)" http://localhost:8080/task 2>/dev/null &
```

done

For HTTPS:

```
for i in {1..20}; do
```

```
curl -k -s -o /dev/null -w "%{http_code}\n" -H "Authorization: Bearer $(python3  
get_token.py)" https://localhost/task 2>/dev/null &
```

done

```
wait | sort | uniq -c
```

[illegible]

8. Failover Test (Stop one API)

For HTTP:

- `sudo docker stop api2`

for i in {1..5}; do

```
curl -s -H "Authorization: Bearer $(python3 get_token.py)" http://localhost:8080/task |  
python3 -c "import sys,json; print(json.load(sys.stdin).get('handled_by'))"
```

done

- `sudo docker start api2`

For HTTPS:

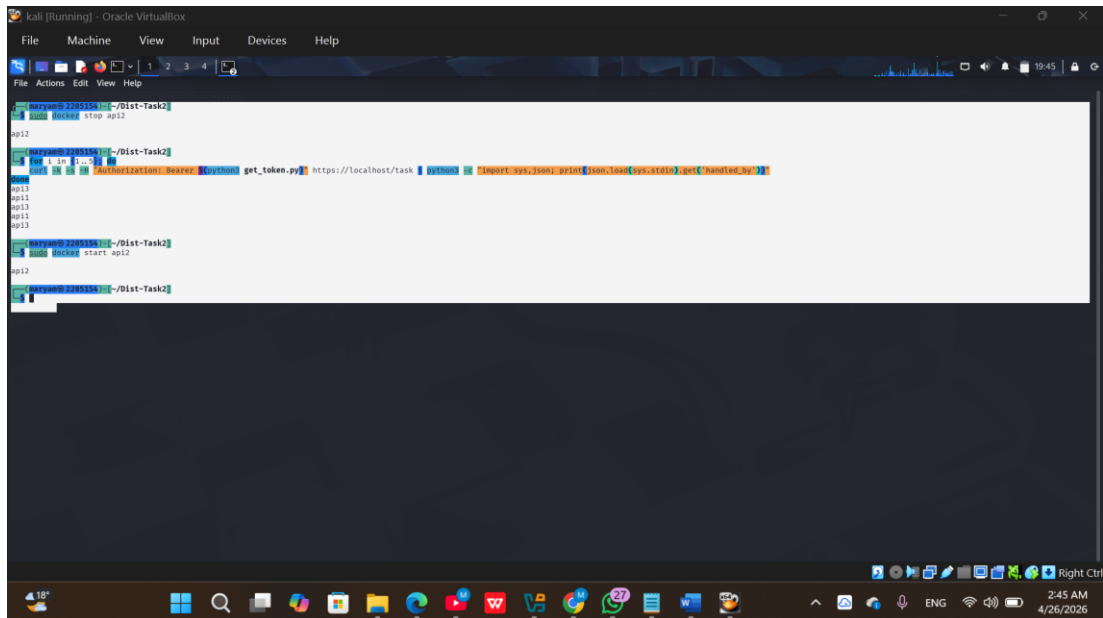
- `sudo docker stop api2`

for i in {1..5}; do

```
curl -k -s -H "Authorization: Bearer $(python3 get_token.py)" https://localhost/task | python3  
-c "import sys,json; print(json.load(sys.stdin).get('handled_by'))"
```

done

- `sudo docker start api2`

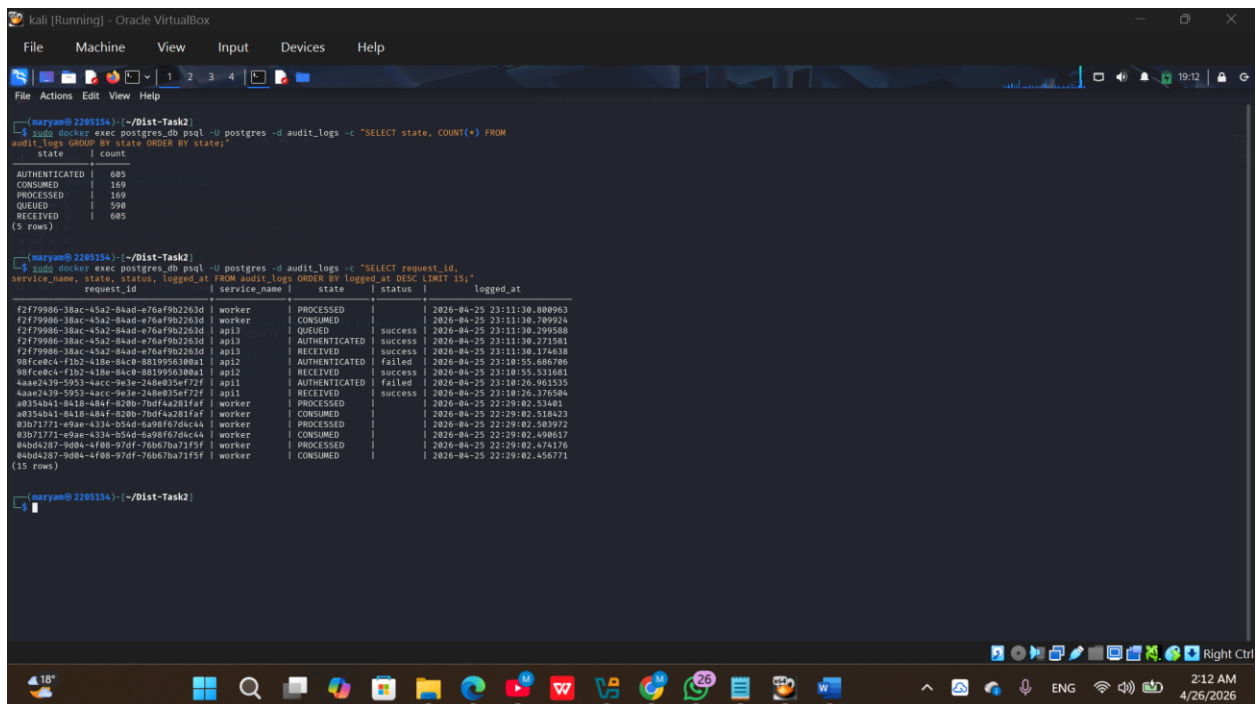


9. DATABASE AUDIT LOGS - All States (RECEIVED, AUTHENTICATED, QUEUED, CONSUMED, PROCESSED)

sudo docker exec postgres_db psql -U postgres -d audit_logs -c "SELECT state, COUNT(*) FROM audit_logs GROUP BY state ORDER BY state;"

Show full audit log content

sudo docker exec postgres_db psql -U postgres -d audit_logs -c "SELECT request_id, service_name, state, status, logged_at FROM audit_logs ORDER BY logged_at DESC LIMIT 15;"



The screenshot shows a Kali Linux terminal window with the following content:

```
(maryam@2205154) (~/.Dist-Task2)
$ sudo docker exec postgres_db psql -U postgres -d audit_logs -c "SELECT state, COUNT(*) FROM audit_logs GROUP BY state ORDER BY state;"
state | count
-----+-----
AUTHENTICATED | 685
CONSUMED | 169
PROCESSED | 169
QUEUED | 598
RECEIVED | 685
(5 rows)

(maryam@2205154) (~/.Dist-Task2)
$ sudo docker exec postgres_db psql -U postgres -d audit_logs -c "SELECT request_id, service_name, state, status, logged_at FROM audit_logs ORDER BY logged_at DESC LIMIT 15;"
request_id | service_name | state | status | logged_at
-----+-----+-----+-----+-----
f2f7998b-38ac-45a2-8aad-e76af9b2263d | worker | PROCESSED | | 2026-04-25 23:11:30.800963
f2f7998b-38ac-45a2-8aad-e76af9b2263d | worker | CONSUMED | | 2026-04-25 23:11:30.789924
f2f7998b-38ac-45a2-8aad-e76af9b2263d | ap13 | QUEUED | success | 2026-04-25 23:11:30.296588
f2f7998b-38ac-45a2-8aad-e76af9b2263d | ap13 | AUTHENTICATED | success | 2026-04-25 23:11:30.271581
f2f7998b-38ac-45a2-8aad-e76af9b2263d | ap13 | RECEIVED | success | 2026-04-25 23:11:30.174638
98fcebca-f1b2-418e-84c8-8819956380a1 | ap12 | AUTHENTICATED | failed | 2026-04-25 23:10:55.688786
98fcebca-f1b2-418e-84c8-8819956380a1 | ap12 | RECEIVED | success | 2026-04-25 23:10:55.531681
4aae2439-5953-4acc-9e3e-2d8e835e722f | ap11 | AUTHENTICATED | failed | 2026-04-25 23:10:26.961525
4aae2439-5953-4acc-9e3e-2d8e835e722f | ap11 | RECEIVED | success | 2026-04-25 23:10:26.374594
a0354b41-8418-484f-820b-7bdfaa281faf | worker | PROCESSED | | 2026-04-25 22:29:02.534081
a0354b41-8418-484f-820b-7bdfaa281faf | worker | CONSUMED | | 2026-04-25 22:29:02.518422
03b71771-e9ae-4334-b54d-6a98f67d4c4a | worker | PROCESSED | | 2026-04-25 22:29:02.503972
03b71771-e9ae-4334-b54d-6a98f67d4c4a | worker | CONSUMED | | 2026-04-25 22:29:02.498617
040b4207-9d04-4f08-97df-7db67ba71f5f | worker | PROCESSED | | 2026-04-25 22:29:02.474176
040b4207-9d04-4f08-97df-7db67ba71f5f | worker | CONSUMED | | 2026-04-25 22:29:02.456771
(15 rows)
```

10. Worker logs

sudo docker start worker

sudo docker logs worker --tail 15

```
kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

# Check worker logs
sleep 5
sudo docker logs worker --tail 20
3eb3077abd91 rabbitmq:management "docker-entrypoint.s..." 9 days ago Up 6 minutes 4369/tcp, 5671/tcp, 0.0.0.0:15672->5672/tcp, :::5672->5672/tcp, 15671/tcp, 15691-15692/tcp, 25672/tcp, 0.0.0.0:15672->15672/tcp, :::15672->15672/tcp
[+] Running 1/1
  ✓ Container rabbitmq Running
  ✓ Container worker Started
    Connection attempt 8 failed:
    Connection attempt 9 failed:
    Connection attempt 10 failed:
    Failed to connect to RabbitMQ
    Worker starting...
    [*] Worker started. Waiting for messages...

Received: b1962e68 from api1
DB: CONSUMED for b1962e68
✓ CONSUMED: b1962e68
DB: PROCESSED for b1962e68
✓ PROCESSED: b1962e68

Received: 2156a7c5 from api2
DB: CONSUMED for 2156a7c5
✓ CONSUMED: 2156a7c5
DB: PROCESSED for 2156a7c5
✓ PROCESSED: 2156a7c5

Received: bd27f55a from api3
[*] Worker started. Waiting for messages...

Received: 0084c3e5 from api1
DB: CONSUMED for 0084c3e5
✓ CONSUMED: 0084c3e5
DB: PROCESSED for 0084c3e5
✓ PROCESSED: 0084c3e5

Received: 61000e2e from api3
DB: CONSUMED for 61000e2e
✓ CONSUMED: 61000e2e
DB: PROCESSED for 61000e2e
✓ PROCESSED: 61000e2e

Received: eaa8ddc0 from api1
DB: CONSUMED for eaa8ddc0
✓ CONSUMED: eaa8ddc0
DB: PROCESSED for eaa8ddc0
✓ PROCESSED: eaa8ddc0

maryam@2285154:~/Dist-Task2
```

11. Containers

```
kali [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

maryam@2285154:~/Dist-Task2
$ sudo docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
d453dc05373    dist-task2-worker      "python -u worker.py"   2 hours ago   Up 29 minutes
a7033ae96eef    dist-task2-api3        "uvicorn app:app --h..." 9 days ago   Up 35 minutes   8080/tcp
09f58048451e    dist-task2-api2        "uvicorn app:app --h..." 9 days ago   Up 24 minutes   8080/tcp
2c8196a7159    dist-task2-api1        "uvicorn app:app --h..." 9 days ago   Up 35 minutes   8080/tcp
f20c3a59b38    nginx:latest           "/docker-entrypoint..." 9 days ago   Up 35 minutes   0.0.0.0:443->443/tcp, :::443->443/tcp, 0.0.0.0:8080->80/tcp, :::8080->80/tcp
2c4673106c75    postgres:15-alpine     "docker-entrypoint.s..." 9 days ago   Up 36 minutes (healthy)   5432/tcp
3eb3077abd91    rabbitmq:management    "docker-entrypoint.s..." 9 days ago   Up 36 minutes   4369/tcp, 5671/tcp, 0.0.0.0:15672->5672/tcp, :::5672->5672/tcp, 15671/tcp, 15691-15692/tcp, 25672/tcp, 0.0.0.0:15672->15672/tcp, :::15672->15672/tcp

maryam@2285154:~/Dist-Task2
```

MITM Attack Simulation (Wireshark)

With HTTP

The image displays a Wireshark network traffic capture on a Kali Linux virtual machine. The interface shows a list of captured packets, with the selected packet (No. 98) expanded to show its details. The packet is an HTTP GET request from 127.0.0.1 to 127.0.0.1, containing a Basic authentication header. The details pane shows the following information:

- Frame 98: Packet, 479 bytes on wire (3832 bits), 479 bytes captured (3832 bits) on interface lo, id 0
- Ethernet II, Src: 00:00:00:00:00:00 (00:00:00:00:00:00), Dst: 00:00:00:00:00:00 (00:00:00:00:00:00)
- Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
- Transmission Control Protocol, Src Port: 51774, Dst Port: 15672, Seq: 919, Ack: 2399, Len: 425
- Hypertext Transfer Protocol
 - GET /api/nodes HTTP/1.1
 - Host: localhost:15672
 - User-Agent: Mozilla/5.0 (X11; Linux x86_64; rv:140.0) Gecko/20180813 Firefox/140.0
 - Accept: */*
 - Accept-Language: en-US,en;q=0.5
 - Accept-Encoding: gzip, deflate, br, zstd
 - Authorization: Basic ZWlic3Q6Z3Vlc3Q=
 - Connection: keep-alive
 - Referer: http://localhost:15672/
 - Cookie: #=5959:true
 - Sec-Fetch-Dest: empty
 - Sec-Fetch-Mode: cors
 - Sec-Fetch-Site: same-origin

The packet details pane also shows the raw packet data in hexadecimal and ASCII format. The ASCII format shows the Basic authentication header: Basic ZWlic3Q6Z3Vlc3Q=.

With HTTPS

kali-linux-2023.4-virtualbox-amd64 [Running] - Oracle VM VirtualBox

File Machine View Input Devices Help

*Loopback: lo

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

tls.stream eq 0

No.	Time	Source	Destination	Protocol	Length	Info
24	5.436247622	:::1	:::1	TLSv1.3	1769	Client Hello (SHA1:localhost)
25	5.43625009	:::1	:::1	TLSv1.3	1093	Server Hello, Change Cipher Spec, Application Data, Application Data, Application Data, Application Data
38	5.453764051	:::1	:::1	TLSv1.3	154	Change Cipher Spec, Application Data
40	5.461634508	:::1	:::1	TLSv1.3	648	Application Data, Application Data

Length: 42
Encrypted Application Data: 8a03af07f55c0afc0e027e09cccf3ef12e032f03e54b41a2334b4005f28365acf9b6f
[Application Data Protocol: Hypertext Transfer Protocol]
- TLSv1.3 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
Opaque Type: Application Data (23)
Version: TLS 1.2 (0x0303)
Length: 992
Encrypted Application Data [..]: 950b2249bba36dd48a15abd613f2aaccae4c15835c5da38d90e7c1c53b15417d
[Application Data Protocol: Hypertext Transfer Protocol]
- TLSv1.3 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
Opaque Type: Application Data (23)
Version: TLS 1.2 (0x0303)
Length: 281
Encrypted Application Data [..]: 906eb2220c0ba9fbc7c83f210cdda37bdc3515acfa1d2513c1c43eaca2c2c7c3f
[Application Data Protocol: Hypertext Transfer Protocol]
- TLSv1.3 Record Layer: Application Data Protocol: Hypertext Transfer Protocol
Opaque Type: Application Data (23)
Version: TLS 1.2 (0x0303)
Length: 69
Encrypted Application Data: b16a2be06c761822473eedc74448d7594f5938ffcc143f65a5b099a1260812cbdf1312d
[Application Data Protocol: Hypertext Transfer Protocol]

Bytes 169-1288: Key Exchange (tls.handshake.extensions_key_share_key_exchange)

Packets: 179 - Displayed: 4 (2.2%) - Dropped: 0 (0.0%)

Profile: Default

22°C
11:02 PM
25/04/2026